

RACHIT KUMAR SINGH

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Education

Kalinga Institute of Industrial Technology, Bhubaneswar, Odisha	JUL 2023 – MAY 2027
B.Tech in Computer Science and Engineering CGPA: 8.31/10 No Active Backlogs	
Vision International School, CISCE Board, Uttarpara Kotrung, West Bengal	
Class XII: 84.4%	APR 2023
Class X: 89.8%	APR 2021

Technical & Soft Skills

Languages:	Python, Java, JavaScript, TypeScript, SQL
CS Fundamentals:	Data Structures & Algorithms, Object-Oriented Design, Complexity Analysis
Systems:	Operating Systems, Distributed Systems, Scalability, Fault Tolerance, Concurrency
Backend & Databases:	FastAPI, REST APIs, JWT, SQLAlchemy, Relational Databases (PostgreSQL, SQLite)
Testing & Security:	pytest, Vitest, Locust (Load Testing), Dependency Vulnerability Scanning
Cloud & DevOps:	AWS EC2, Docker, GitHub Actions CI/CD, Git, JIRA, Prometheus, Grafana
Soft Skills:	Ownership, Problem Solving, Written & Verbal Communication, Root-Cause Analysis

Work Experience

Accenture – Associate Software Engineering Intern, Bengaluru MAY – JUL 2026

Project: Specification Driven Development (SDD) in Application Management Services (AMS)

- Selected as 1 of 9 from 28 interns for a core project track; co-owned SDD in AMS with one teammate end to end
- Built an AI-powered RAG support chatbot with FastAPI that replaced manual L1 ticket triage
- Diagnosed 3 recurring root causes in AMS ticket workflows (requirement ambiguity, documentation gaps, traceability issues); designed automated root-cause analysis with confidence scoring over a resolved-ticket knowledge base
- Modeled L1-to-L2/L3 escalation (13 evaluation criteria, 15 process rules), automating hand-offs across 4 teams
- Authored Software Design Documents (SDDs) piloted across 2 delivery tracks, built a JIRA-based demo prototype, and triaged support tickets across priority levels within SLAs in Agile sprints

Technical Projects

APIWatch

[GitHub](#)

Python, FastAPI, SQLAlchemy, PostgreSQL, SQLite, pytest, GitHub Actions, Prometheus, Grafana

- Designed a FastAPI REST API with JWT authentication (bcrypt-hashed passwords) and per-user data isolation
- Prevented lost updates with optimistic concurrency control (version field; stale writes return 409 Conflict) and made writes retry-safe via Idempotency-Key handling
- Engineered a 49-test suite at 98.63% coverage, enforced by a GitHub Actions CI pipeline gating merges on lint (ruff/black) and a coverage threshold
- Instrumented Prometheus metrics and a Grafana dashboard (latency, throughput, error rate) under concurrent load
- Added /health/live and /health/ready probes; verified SQLite and PostgreSQL 16 support with zero code changes

Smart Policy Recommendation Engine

[GitHub](#)

React, TypeScript, FastAPI, SQLAlchemy, Alembic, scikit-learn, Docker, GitHub Actions, pytest, Vitest

- Built a multi-tiered recommendation engine (React + TypeScript, async FastAPI + SQLAlchemy) ranking explainable policies via a hard-eligibility filter, rule-based scoring, and ML pricing
- Trained a scikit-learn Gradient Boosting Regressor on a 1,338-row dataset (R-squared 0.87 on a held-out test set), blended with the rules engine as a transparent pricing signal
- Achieved 95%+ coverage across 45 automated tests (pytest, Vitest + React Testing Library) with CI coverage gates, dependency vulnerability scanning, and Docker builds
- Containerized 4 services with Docker Compose and load-tested the API with Locust at 50 concurrent users

Certifications, Achievements & Activities

- Solved 300+ DSA and SQL problems on LeetCode (60+ medium, 20+ hard)
- SOF IMO: 3 Gold Medals (2014–2016) and 2 Silver Medals (2017–2018) across five national cycles
- Basketball – XIM Open Tournament: finished top 16 among 100+ registered teams
- IEEE CTSoc & CIS Technical Volunteer – 23rd OCIT International Conference, KIIT

Accenture Summer Internship Badge and Certification

SAP Certified – SAP Fiori Application Developer